Problem 02: Multimodal Deepfake Detection and Media Forensics Platform

Domain: Cybersecurity | Computer Vision | Audio Processing | Multimodal AI
Difficulty: Hard

Problem

Deepfakes (fake video, cloned voice, manipulated images) enable fraud, impersonation and misinformation. Single-modality detectors fail when attackers combine fake video with synthetic audio, or when compression hides artifacts.

Objective

Build a media-verification platform that accepts image, video, audio or combinations and returns a structured forensic assessment with explanations.

Users

Journalists, fact-checkers, bank fraud teams, trust-and-safety analysts.

Core Requirements

1. Image, video and audio upload.
2. Automatic frame and audio-segment extraction.
3. Detect facial artifacts and visual inconsistencies.
4. Detect synthetic speech / voice manipulation.
5. Temporal consistency analysis across frames.
6. Lip-sync check between lips and audio.
7. Multimodal fusion of modality predictions.
8. Separate image, video and audio risk scores.
9. Highlight suspicious frames and audio segments.
10. Forensic report with evidence and limitations.

Advanced Challenges

- Generalization to unseen generators (train on one, test on another).
- Robustness to compression, resizing, noise, re-encoding.
- Cross-modal audio-video consistency.
- Confidence calibration.
- Partially manipulated media (localized edits).

Suggested Datasets

FaceForensics++, Celeb-DF v2, DFDC (sample), FakeAVCeleb (audio-video), ASVspoof 2019/2021, WaveFake, In-the-Wild audio deepfakes.

Suggested Tech

PyTorch, OpenCV, ViT / EfficientNet, wav2vec2 or AASIST / spectrogram CNNs, SyncNet-style lip-sync, Grad-CAM, FastAPI.

Reference Architecture

Upload → ffmpeg demux → face detector (e.g., RetinaFace/MediaPipe) → visual branch; audio branch; temporal branch; sync branch → fusion head (late fusion first, attention fusion as extension) → calibrator (temperature scaling) → report generator.

Evaluation

AUC, EER (audio), accuracy per generator (cross-dataset), robustness curves under JPEG/H.264 compression levels, ECE (calibration). Baselines: single-modality models, simple averaging fusion.

Expected Deliverable

App with per-modality scores, flagged frames/segments, heatmaps, confidence, limitations, downloadable PDF report.

Safety Note

Report “evidence of potential manipulation”, never “definitive proof”.

Pitfalls

Leakage from same identity across train/test; reporting in-dataset accuracy only.

Suggested Timeline (48h)

0-10h data + baselines; 10-24h audio and visual models; 24-34h fusion + calibration; 34-42h robustness tests; 42-48h UI + report.

Standard Judging Rubric (applies to every problem)

Criterion	Weight
Problem understanding	10%
Technical depth	20%
Data engineering (quality, leakage prevention, reproducibility)	10%
Model performance (metrics, baselines, honest evaluation)	15%
System architecture	15%

Criterion	Weight
Working prototype	10%
Explainability and responsible AI	10%
Innovation and scalability	5%
Presentation and demo	5%

Mandatory Submission Checklist

- ☐GitHub repository with setup instructions
- ☐Working prototype or recorded demo
- ☐Documented system architecture
- ☐Dataset sources and preprocessing methodology
- ☐Evaluation results with baseline comparisons
- ☐Limitations and failure cases
- ☐One technically challenging feature demonstrated
- ☐Short presentation: problem, approach, results, future work